INCH-POUND

MIL-PRF-6799M
17 September 2025
SUPERSEDING
MIL-PRF-6799L
w/AMENDMENT 1
14 September 2020

PERFORMANCE SPECIFICATION

COATINGS, SPRAYABLE, STRIPPABLE, PROTECTIVE, WATER EMULSION

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification establishes the performance requirements for water emulsion, protective, strippable, sprayable, coatings for application over metallic, painted and plastic surfaces. The coating consists of a black non-chromate basecoat and white or olive drab non-methanol topcoat (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

Comments, suggestions, or questions on this document should be addressed to Commanding Officer, Naval Air Warfare Center Aircraft Division Lakehurst, System Standardization and PHS&T Branch, Code BL32600, Mail Stop B120-3, Route 547, Joint Base MDL, NJ 08733-5100 or emailed to NAVAIR-Standardization@us.navy.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

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2.2 Government documents.

2.2.1 Specifications and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

FEDERAL SPECIFICATIONS

P-P-560	-	Polish, Plastic.
TT-N-95	-	Naphtha, Aliphatic.

COMMERCIAL ITEM DESCRIPTION

A-A-857	-	Thinner, Dope and Lacquer (Cellulose Nitrate).

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-PRF-5425	-	Plastic Sheet, Acrylic, Heat Resistant.
MIL-C-8514	-	Coating Compound, Metal Pretreatment, Resin-acid.
MIL-PRF-22750	-	Coating, Epoxy, High-Solids.
MIL-PRF-23377	-	Primer Coatings: Epoxy, High Solids.
MIL-PRF-81352	-	Coating, Aircraft Touch-up.
MIL-DTL-81772	-	Thinner, Aircraft Coating.
MIL-PRF-85285	-	Topcoat, Aircraft and Support Equipment

(Copies of these documents are available online at https://quicksearch.dla.mil.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

CODE OF FEDERAL REGULATIONS DEPARTMENT OF TRANSPORTATION

Title 49	-	Transportation, Parts 100-199

(Copies of these documents are available online at https://www.ecfr.gov.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

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AMERICAN SOCIETY FOR QUALITY (ASQ)

ASQ-Z1.4	-	Sampling Procedures and Tables for Inspection by Attributes.

(Copies of this document are available from https://asq.org.)

ASTM INTERNATIONAL

ASTM-D2369	-	Coatings, Volatile Content of
ASTM-D6695	-	Standard Practice for Xenon-Arc Exposures of Paint and Related Coatings

(Copies of these documents are available from https://www.astm.org.)

SAE INTERNATIONAL

SAE AMS-QQ-A-250/5	-	Aluminum Alloy, Alclad 2024, Plate and Sheet
SAE AMS-STD-595	-	Color number 36440 – Gray, Flat or Lusterless.
SAE AMS-STD-595	-	Color number 24084 – Green, Semigloss.

(Copies of these documents are available from https://www.sae.org/.)

2.3 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Qualification. The coatings furnished under this specification shall be products that are authorized by the qualifying activity for listing on the applicable qualified products list before contract award (see 4.2 and 6.3).

3.2 Materials. The coatings shall be made from materials that ensure compliance with the requirements of this specification.

3.2.1 Form.

3.2.1.1 System coatings. Coatings shall be water emulsions which are capable of being sprayed and stripped as a system. Coatings shall be a two-part coating system consisting of a black pigmented basecoat and a compatible white pigmented topcoat. When specified (see 6.2), olive drab pigmented topcoat is an alternate.

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3.2.2 Toxicity. The coatings shall have no adverse effect on the health of personnel when used for its intended purpose (see 4.5.18, 6.4 and 6.5).

3.3 Performance.

3.3.1 Properties. The physical, mechanical, and chemical properties of the individual components and the coating systems shall meet the performance requirements specified in tables I and II when tested in accordance with the specified test methods.

3.3.2 Apparent viscosity. Apparent viscosity of the basecoat and topcoat shall be capable of being delivered using the spray application equipment while meeting all performance requirements of this specification. Viscosity shall be measured to ensure product consistency for qualification and conformance testing (see 4.5.15).

TABLE I. Individual component physical properties.

Physical Property	Requirement		Test Method Paragraph
	Basecoat	Topcoat	
Chemical attack	No attack on plastic	No attack on plastic	4.5.1
Nonvolatile content, min., % by weight	37	48	4.5.2
Delivery rate, cc/min	175-700	250-700	4.5.3
Vertical application(wet)	No sagging 12-14 mils thickness	No sagging 16-20 mils thickness	4.5.4
Drying time, max.	2 hrs.	2 hrs.	4.5.5
Storage	1/	1/	4.5.13

1/ After storage exposure, coating shall meet the requirements of delivery rate, vertical application, drying time, apparent viscosity, adhesion (initial only), tensile strength and elongation tests.

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TABLE II. Coating system properties.

Property	Requirement	Test Method Paragraph
Craze attack on plastic	Shall protect the surface of the acrylic plastic from attack and shall be strippable after exposure.	4.5.6
Adhesion, initial (as received)	3-20 oz/inch, width	4.5.7
Adhesion after accelerated drying	8-50 oz/inch, width	4.5.8
Adhesion after exposure to high humidity	8-50 oz/inch, width, no dissolution of coating	4.5.9
Tensile strength	1700 psi, min.	4.5.10
Ultimate elongation	170-350 percent	4.5.10
Stability	Shall meet the requirements of the following tests after exposure to the stability test cycle: a. Strippability after accelerated weathering. b. Adhesion (all conditions) c. Tensile strength d. Ultimate elongation	4.5.11
Weathering	Coating shall not show checking, cracking, or embrittlement which penetrates to the substrate, and shall be strippable in one continuous sheet. There shall be no crazing of plastic surfaces. Finished surfaces shall not lift, mar, or show other irregularities. Aluminum panels shall not corrode.	4.5.12
Water erosion resistance	Coating shall not lift, check, or erode.	4.5.14

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3.3.3 Condition in container. When tested as specified in 4.5.16, the coating shall be free from grit, seeds, skins, lumps, and livering and shall show no pigment settling or caking that cannot be returned to a homogeneous state by agitating/mixing as recommended by the manufacturer.

3.3.4 Field evaluation. Manufacturers shall submit a 5-gallon pail of black basecoat and a 5-gallon pail of white topcoat to AMARG (Aerospace Maintenance and Regeneration Group) to perform field testing of submitted product using their normal production spray application equipment. Coatings will be evaluated for sprayability and performance after initial drying and after 30 days. When tested in accordance with 4.5.17 the coating shall show no checking, cracking, or embrittlement which penetrates to the substrate, and shall be strippable in one continuous sheet. There shall be no crazing of plastic surfaces. Finished surfaces shall not lift, mar, or show other irregularities.

3.4 Precautionary marking. Each container shall be marked with the following precautionary marking in letters not less than 0.75 inch high.

"DO NOT USE IF ANY INDICATION OF FREEZING" AND
"MIX BEFORE USE AS RECOMMENDED BY MANUFACTURER"

3.5 Freeze indicators. Freeze indicators shall be supplied as part of each container of basecoat and topcoat.

3.6 Workmanship. Each component shall be filtered prior to packaging resulting in homogenous quality, consistency, and free of agglomerates and foreign particles.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2).

b. Conformance inspection (see 4.3).

4.2 Qualification inspection. Qualification inspection of the coatings shall consist of all tests and examinations of this specification.

4.3 Conformance inspection. Conformance inspection of the coatings shall consist of the examinations and tests specified in table III and table IV.

4.3.1 Sampling.

4.3.1.1 Sampling for tests. Two containers of basecoat and topcoat shall be randomly selected from each lot offered for inspection. An amount of material sufficient to perform the

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tests specified in 4.3.2.1 shall be withdrawn from each of the two containers and placed in separate clean, dry, metal, or glass containers. Each container shall be tested as specified in 4.3.2.1.

4.3.1.2 Examination of end item. A random sample of filled containers shall be selected from each inspection lot in accordance with ASQ-Z1.4, Inspection Level I, for examination of the end item as specified in 4.3.2.2.

4.3.2 Inspection and tests.

4.3.2.1 Testing. The samples selected in 4.3.1.1 shall be subjected to the tests specified in table III. Failure of any sample to pass all the tests shall be cause for rejection. The lot represented by the test samples shall be rejected.

TABLE III. Conformance tests.

Property	Requirement	Test Method Paragraph
Nonvolatile content	Table I	4.5.2
Delivery rate	Table I	4.5.3
Vertical application	Table I	4.5.4
Adhesion, initial (as received)	Table II	4.5.7
Apparent viscosity	3.3.2	4.5.15

4.3.2.2 Visual examination. The sample selected in accordance with 4.3.1.1 shall be visually examined for the defects specified in table IV. There shall be no defects.

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TABLE IV. Visual examination of the end item.

Examine	Defect
Form	Not as specified.
Appearance	Color not as specified. Coating filled with lumps. Foreign matter present.
Fill of individual coating container (metal can, pail, drum)	Quantity of coating not as specified.
Condition in container	Grit, seeds, skins, lumps, livering, pigment settling or caking.
Freeze indicators	Not supplied as part of each container. Indicate freezing has occurred.
Marking	Not as specified 3.4.
Workmanship	Not having consistency nor homogenous quality, free of agglomerates and foreign particles

4.4 Inspection conditions. Unless otherwise specified, all tests shall be conducted at a temperature of 70 ± 5 ˚F and 50 ± 10 percent relative humidity.

4.4.1 Plastic test panels. All plastic test panels shall be made from acrylic plastic sheet conforming to MIL-PRF-5425, Finish A. The panel shall be given a preliminary cleaning with a cloth saturated with water or aliphatic naphtha conforming to type II of TT-N-95, rinsing with warm water, and drying with a lint-free cloth. Polish conforming to P-P-560 Type I shall be evenly applied, allowed to dry, and wiped clean with a lint-free cloth.

4.4.2 Aluminum test panels. Aluminum panels shall be made from aluminum alloy (Alclad 2024) conforming to SAE-AMS-QQ-A-250/5.

4.4.3 Coating of panels. Plastic and aluminum panels shall be coated.

4.4.3.1 Two-coat system. The two coat system shall consist of a combination of basecoat with a topcoat.

4.4.3.1.1 Basecoat with topcoat system. Basecoat material shall be applied in one horizontal pass and one vertical pass until a wet film thickness of 12-14 mils is achieved. Allow to air dry for one hour and apply a second coat of basecoat material and allow to dry for a minimum of 2 hours at 73 ± 3 °F. This procedure shall produce a total basecoat dry film thickness of 12-14 mils. The topcoat material shall then be applied over the basecoat

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material. Topcoat material shall be applied in one horizontal pass and one vertical pass until a wet film thickness of 16-20 mils is achieved. Allow to air dry for one hour and apply a second coat of topcoat material and allow to dry for a minimum of 2 hours at 73 ± 3 °F. This procedure shall produce a total topcoat dry film thickness of 16-20 mils.

4.4.4 Mechanical mixing. All samples to be tested in 4.5 shall be agitated/mixed as recommended by the manufacturer.

4.5 Test methods.

4.5.1 Chemical attack. Two 1 x 4 x 0.125 inch plastic strips shall be cleaned as specified in 4.4.1. One strip shall be immersed to one-half of its length in a screw-top bottle containing the basecoat. One strip shall be immersed to one-half of their length in a screw-top bottle containing the topcoat. Both lids shall be tightened, and all bottles stored for 24 hours at a temperature of 100 ± 3 °F. The strips shall then be removed from the bottles and examined for crazing, cracking, or discoloration on the plastic. The coatings shall meet the requirement specified in table I.

4.5.2 Nonvolatile content. The nonvolatile content shall be determined in accordance with ASTM D2369 and meet the requirement specified in table I.

4.5.3 Delivery rate. A conventional spray gun using a 1.7-1.8mm (0.067– 0.070 inch) orifice fluid tip and a maximum fluid-nozzle screw adjustment shall be used. The line pressure shall be 75-100 psi. The volume of coating delivered in 1 minute shall be noted. The delivery rate shall meet the requirements specified in table I.

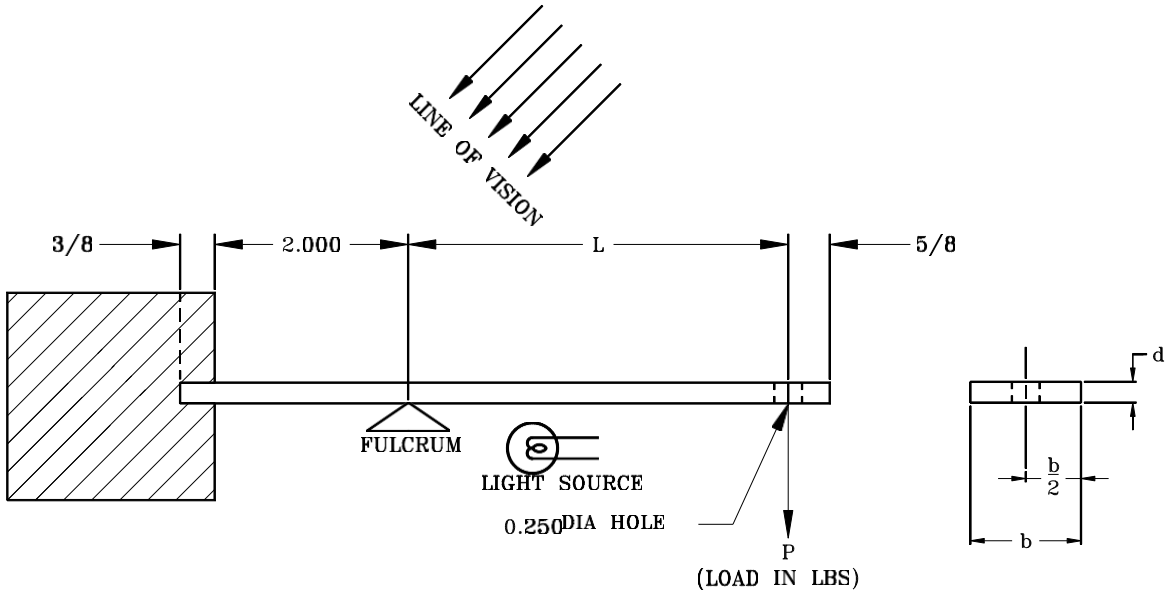
4.5.4 Vertical application. The coating shall be placed in a pressure feed cup and sprayed with a gun described in 4.5.3 on a vertical plastic panel. The gun shall be held perpendicular to and 8 to 10 inches from the panel and shall be adjusted so that one cross hatched application of each coating produces the required wet thickness (see table I). The panel shall be examined for sagging of the coating and shall meet the requirements specified in table I.

4.5.5 Drying time. Two 4 x 8 x 0.250 inch plastic panels shall be coated individually as follows: Basecoat shall be crosshatched coated to a wet thickness of 12-14 mils; Topcoat shall be crosshatched coated to a wet thickness of 16-20 mils. The coated panels shall be conditioned at a temperature of 70 ± 5 ˚F and 50 ± 10 percent relative humidity for the required drying time (see table I). After each application, the coating shall be dry to touch within the maximum time specified in table I. Two applications are required for each coating.

4.5.6 Craze attack on plastic for basecoat and topcoat. Six machined plastic panels, each 1 x 7 x 0.250 inch, cleaned as specified in 4.4.1, shall be set up as cantilever beams as shown on figure 1, and stressed to 2,400 psi. The formula for calculating load shall be as indicated in figure 1. Ten minutes after the beams have been loaded and while still under load they shall be examined for crazing. Three panels under stress shall then be coated on

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the top side by applying a puddle of the coating to the stressed plastic, taking care not to expose the side of the specimen, and shall be allowed to air dry for 24 hours. Three uncoated panels shall be used as a control. At the end of the exposure time, the stress shall be removed, the coatings stripped, and all panels examined for crazing. All panels shall meet the requirements specified in table II. In the event the control panels craze, tests shall be repeated. All craze appearing at the edges of the specimen shall be disregarded unless they extend 0.125 inch or more from either edge of the specimen.



DIMENSIONS IN INCHES. TOLERANCES: DECIMALS, ±0.005, FRACTIONS, ±1/64

Formula for calculating load: $P = \frac{Sbd^2}{6L}$

Where:

S = Stress in pounds per square inch
P = Load in pounds
L = Distance from fulcrum to load in inches
b = Width of panel in inches (measured to nearest 0.001)
d = Thickness of panel in inches (measured to nearest 0.001)

NOTE: To prevent heating of specimens, the light source shall be utilized only when examining the specimen for craze.

FIGURE 1. Typical stress crazing beam.

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4.5.7 Adhesion, initial (as received). Five bare plastic panels, and five aluminum panels painted in accordance with in 4.5.12.1e, each 1.5 x 6 x 0.125 inches, shall be coated as specified in 4.4.3. The panels shall then be allowed to air dry 24 hours prior to testing. A one-inch-wide strip shall be cut only through the coating and parallel to the long dimension of the panel (allowing 0.25 inch on each side). The panel shall be secured to the peel test fixture of the tensile testing machine. One end of the 1 inch strip of coating shall be loosened from the panel (not more than 1 inch) and attached to the upper clamp. The load shall be applied at an angle of 90°. The testing machine shall have at least a 4-pound range, and the rate of jaw separation shall be 12 inches per minute. The load required to remove the strip of coating from the panel shall be recorded for each panel and the average load in ounces per inch of width reported for each type of panel. The results shall be recorded. The coating adhesion shall meet the requirement as specified in table II.

4.5.8 Adhesion after accelerated drying. Five bare plastic panels, and five aluminum panels painted in accordance with in 4.5.12.1e, each 1.5 x 6 x 0.125 inches, shall be coated as specified in 4.4.3, and allowed to air dry 2 hours prior to testing and then subjected for 100 hours to accelerated drying by exposure in a desiccator (containing fresh anhydrous calcium chloride) at a temperature of 100 ± 5 °F. The coating shall be removed from each panel as specified in 4.5.7. The load required to remove the strip of coating from the panel shall be recorded for each panel and the average load in ounces per inch of width reported for each type of panel. No residue shall be left on the surface of the plastic or painted aluminum. The coating adhesion shall meet the requirement specified in table II.

4.5.9 Adhesion after exposure to high humidity. Five bare plastic panels, and five aluminum panels painted in accordance with in 4.5.12.1e, each 1.5 x 6 x 0.125 inches, two hours after coating as in 4.4.3, shall be subjected for 24 hours to an atmosphere of 100 percent relative humidity at a temperature of 120 ± 5 °F. The panels shall be air dried for 24 hours at room temperature. The coating shall be removed from each panel as specified in 4.5.7. The load required to remove the strip of coating from the panel shall be recorded for each panel and the average load in ounces per inch of width reported for each type of panel. The coating adhesion shall meet the requirement specified in table II.

4.5.10 Tensile strength and ultimate elongation. Coating as specified in 4.4.3 shall be sprayed on an 8 x 8 inch glass panel or UHMW plastic and air dried for 24 hours, removed carefully, and cut into five 1 x 6 inch specimens. The specimens shall be oven dried at a temperature of 120 ± 5 °F for 24 hours. The specimens shall then be conditioned for 4 hours at a temperature of 75 ± 5 °F and a relative humidity of 50 ± 5 percent. The thickness of the film shall then be measured within the gage length using a dial indicator having an accuracy of 0.001 inch, with a 0.25 inch diameter flat foot, under a total load of 1.2 ± 0.1 ounces. A tensile testing machine shall be used, and the rate of jaw separation shall be 12 inches per minute. The average tensile strength and ultimate elongation shall be recorded and shall meet the requirement specified in table II.

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4.5.11 Stability. One gallon of the coating shall be placed in a container of the same type (except for size and method of closure) as normally provided by the supplier, the lid tightened, and the container subjected to the following successive conditions:

a. 120 hours at a temperature of 45 ± 5 °F.
b. 24 hours at room temperature.
c. 120 hours at a temperature of 120 ± 5 °F.
d. 24 hours at room temperature.

NOTE: Test shall be started on Thursday or Friday of a standard work week to avoid material changeover during the weekend.

4.5.11.1 Stability Performance. Panels coated as specified in 4.4.3 with the coating tested above shall then be subjected to and pass the following tests. If the coating is too viscous, enough water shall be added to obtain the recommended delivery rate.

a. Adhesion, initial (as received).
b. Adhesion after accelerated drying.
c. Adhesion after exposure to high humidity.
d. Tensile strength and ultimate elongation.

4.5.12 Weathering.

4.5.12.1 Preparation of panels. Three-inch by six inch panels, prepared as shown herein, shall be coated as specified in 4.4.3.

a. Three bare panels of 0.125 inch thick plastic conforming to MIL-PRF-5425 and prepared as specified in 4.4.1.

b. Three bare panels of aluminum alloy (Alclad 2024) conforming to SAE-AMS-QQ-A- 250/5.

c. Three aluminum panels sprayed with one coat of epoxy-polyamide conforming to MIL-PRF-23377 to a dry film thickness of 0.6 mils to 0.8 mils and air dried for three hours. Panels shall then be sprayed with two cross coats of lacquer conforming to MIL-PRF-81352 to a total dry film thickness of 1.2 ± 0.2 mils, with a one-half hour air dry interval between coats. The lacquer shall have been reduced 1 volume package material to 1.5 volumes thinner composed of 2 parts toluene to one part xylol by volume. Air dry panels for 1 week before testing.

d. Three aluminum panels sprayed with one coat of wash primer conforming to MIL-C-8514 to a dry film thickness of 0.2 mils to 0.3 mils and air dried for 30 minutes. Panels shall then be coated as specified in 4.5.12.1c.

e. Three aluminum panels sprayed with one coat of epoxy-polyamide primer conforming

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to MIL-PRF-23377 to a dry film thickness of 0.6 mils to 0.8 mils and air dried for three hours. Panels shall then be sprayed with two cross coats of polyurethane coating, conforming to MIL-PRF-85285 (color number 36440 of SAE AMS-STD-595) to a total dry film thickness of 1.2 ± 0.2 mils with a one-half hour air dry interval between coats. The polyurethane coating material shall have been reduced in accordance with MIL-DTL-81772 thinner to a viscosity of 16-18 seconds (#2 Zahn cup). Air dry panels for one week before testing.

f. Three aluminum panels prepared as follows: Spray the panels with one coat of epoxy primer conforming to MIL-PRF-23377 to a dry film thickness of 0.6 mils to 0.8 mils and air dry for one hour. Apply a mist coat of white epoxy finish conforming to MIL-PRF-22750 and air dry 60 minutes. Then apply a full coat of the white epoxy finish to a dry film thickness of 1.1 mils to 1.5 mils. The white epoxy finish shall have been reduced 2 volumes of admixed epoxy topcoat to 1 volume of A-A-857 thinner. Dry for a minimum of 24 hours.

4.5.12.2 Procedure. After application of the coating, the panels shall remain for at least 24 hours at a temperature of 70 to 80 °F and then be placed in a Xenon Arc apparatus for a period of 1,000 hours in accordance with ASTM D6695 Cycle 1. At the end of the exposure period the coated panels shall be examined for checking, cracking, embrittlement, lifting, peeling, and erosion; coating shall be examined for strippability; plastic surface examined for crazing, haze, or other residue; and the painted surface examined for lifting, marring, staining, or other irregularities. The aluminum panels shall be examined for corrosion. The coating shall meet the requirement specified in table II.

4.5.13 Storage Stability. For qualification testing of storage stability, a one gallon container of both basecoat and topcoat, as packaged and shipped by the manufacturer's regular shipping methods shall be used. The unopened container shall be stored for 1 year in a temperature range that shall be not less than 40 °F and not greater than 90 °F at a relative humidity of 50 ± 10 percent. At the end of the storage period, the container shall be opened and visually inspected. The coating shall not be permanently gelled, nor shall it show indication of livering or separation. If successful, the coating shall then be mixed as recommended by the manufacturer and tested to meet the requirements specified in table I for delivery rate, vertical application and drying time. All samples shall then be tested in accordance with table II to meet the requirements for viscosity, adhesion (initial only), tensile strength and elongation.

4.5.14 Water erosion resistance. Three bare plastic panels, and three aluminum panels painted in accordance with in 4.5.12.1e, each 3 x 6 x 0.125 inches, shall be coated as specified in 4.4.3. After two hours of air drying, the panels shall be exposed in a horizontal position to simulated rainfall for two hours. The rainfall shall consist of a steady spray of water producing uniform droplets at a rate of 2 ± 0.5 inches per hour. After waiting 24 hours, the panels shall be examined for lifting, checking, or erosion of the coating and shall meet the requirement specified in table II.

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4.5.15 Apparent viscosity. Apparent viscosity shall be measured using a LVT viscometer at 73 ± 3 ºF at 5 revolutions per minute (rpm), shall be 9,000 - 14,000 centipoise (cps) for basecoat and 2,000 - 6,000 centipoise (cps) for topcoat. A number 5 spindle shall be used and immersed to the center of immersion groove. The sample shall be contained in a beaker or cylinder not less than 3.0 cm in diameter to a depth of not less than 4.5 cm. All measurements shall be made with the rotor guard removed from the instrument, and apparent viscosity shall be computed without the application of the correction factors normally applied when the instrument is used without a guard. Samples shall be stirred to ensure uniformity. The temperature of the sample shall be checked before measurements by immersing a thermocouple or thermometer in the sample to ensure that the temperature is at 73 ± 3 °F. The spindle shall be allowed to rotate at 5 rpm until a constant reading is obtained. Pairs of readings at the same speed should not differ by more than 10%, and the apparent viscosity shall be within the ranges specified in 3.3.2. Upon qualification, the manufacturer shall maintain the recorded qualified viscosities within ± 5% for both the basecoat and topcoat of this product.

4.5.16 Condition in container. The coating shall be examined visually for the defects as specified in 3.3.3. All samples to be tested shall be mixed as recommended by the manufacturer to determine compliance with the requirements specified in 3.3.3.

4.5.17 Field evaluation test. AMARG shall perform field testing of submitted product using their normal production spray application equipment. At a minimum, one designated aircraft surface and one canopy section shall be exposed and evaluated for field testing. After application, the coated surfaces shall meet the requirements specified in 3.3.4.

4.5.18 Toxicity. The product shall be evaluated by the Navy and Marine Corps Public Health Center (NMCPHC) using the administrative Health Hazard Assessment (HHA) (see 6.5).

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in- house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful but is not mandatory.)

6.1 Intended use. These coatings are for use exclusively in military applications to cover military aircraft for long term storage in a desert environment and for transport in extreme seagoing environments not encountered by commercial aircraft. They are intended for use as follows:

a. Basecoat, black. This material is intended for use as (1) a strippable, protective coating for acrylic plastic bulk materials and assemblies containing acrylic plastics when the protected item is shipped fully covered or stored under cover, and (2) as a basecoat for the manufacturer's compatible topcoat.

b. Topcoat, white or olive drab. This material is intended to be used only as a topcoat for the manufacturer's compatible basecoat. In combination, this protective coating system serves as a sprayable, strippable, protective coating for application on metallic, painted and plastic surfaces, such as entire aircraft, missiles, rockets, and transportation vehicles during outdoor storage and overseas deck loaded shipments. For Army use, the color should conform to SAE AMS-STD-595, Color No. 24084.

6.1.2 Unintended use. The coatings should not be used as a barrier or intermediate coat under lacquers, primers, enamels, or similar materials.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of the specification.
b. If olive drab color is required for topcoat.
c. Quantity.
d. Name and address of the Government activity responsible for conducting the conformance inspection program.
e. Type and capacity of containers required.
f. Selection of capacity of containers required.
g. Packaging requirements.
h. Special marking, when required.

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Products List QPL-6799, whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products and the letter of authorization for submittal of sample may be

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obtained from the Qualifying Activity: Commander, Naval Air Warfare Center Aircraft Division Lakehurst, System Standardization and PHS&T Branch, Code BL32600, Mail Stop Bldg. 333, Route 547, Joint Base MDL, NJ 08733-5100 or at NAVAIR-Standardization@us.navy.mil. An online listing of products qualified to this specification may be found in the Qualified Products Database (QPD) at https://assist.dla.mil.

6.3.1 Qualification submittal. To initiate the qualification process, forward a written request to submit product, to the qualification activity (see 6.3). A written response to this will be a "letter of authorization," to submit product for testing. The letter will provide detailed instructions for the submission of product (e.g., sample size, sample identification, testing fees, test data and reports, etc.).

6.3.2 Qualification approval. The qualifying activity upon completion of product testing will issue a “letter of notification” either notice of failure with testing progress or notice of product approval with a listing of mandatory manufacturer responsibilities to have their product listed and remain authorized on the Qualified Products List (QPL). As applicable, the approval letter will detail other stipulations of qualification, retention of qualification, QPL listing information, formal date of qualification, etc.

6.3.3 Retention of qualification. In order to retain qualification of a product approved for listing on the Qualified Product List (QPL), the Government will require the manufacturer to verify by certification to the qualifying activity that the manufacturer’s product complies with the requirements of this specification. The time of periodic verification by certification will be in two year intervals from the date of original qualification and will be initiated by the Government. The Government reserves the right to re-examine the qualified product whenever deemed necessary to ensure that the product continues to meet any or all of the specification requirements

6.3.4 Miscellaneous. Coatings supplied under contract is to be identical in every respect to the samples tested and found to meet the requirements of this specification. Any unapproved changes from the qualification sample should constitute cause for rejection for material submitted and for removal from the list of qualified products.

6.4 Toxicological data and formulations. The manufacturer should furnish the toxicological and formulation data to evaluate the safety of the material for the proposed use. A statement of the complete formulation of the product, to include the chemical name and percent of each ingredient, as well as a material safety data sheet of the product, prepared in accordance with FED-STD-313, should be submitted to the qualifying activity. The pertinent government mailing addresses for submission of data are listed in FED-STD-313.

6.5 Health/hazard assessment (HHA). The qualifying activity will request the Navy Environmental Health Center (NAVENVIRHLTHCEN) to perform an HHA on the product furnished. Prior to listing on the Qualified Products List, the results of the NAVENVIRHLTHCEN HHA must be deemed acceptable by the qualifying activity. The HHA is a review of the product based on information submitted by the manufacturer, to assess health hazards associated with the handling, application, and use of the product. Information on the HHA process can be found at:

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http://www-nehc.med.navy.mil/occupational_health/industrial_hygiene/hazmat_ahha.aspx. (BUMEDINST 6270.8B provides additional information/guidance pertaining to this process). Sufficient data to permit an HHA of the product is to be provided by the manufacturer to the Commanding Officer, Navy Environmental Health Center, 620 John Paul Jones Circle, Suite 1100, Portsmouth, VA 20378-2103.

6.6 Repair coatings. Repair or patching should be accomplished using the same manufacturer's compatible basecoat and topcoat.

6.7 Subject term (key word) listing.

Aircraft coating
Aircraft protection
Protection
Desert storage
Multi-coat system
Strippable coating
Removable coating
Outdoor storage
Preservation
Protective coating
Stripper
Corrosion protection

6.8 Changes from the previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

CONCLUDING MATERIAL

Custodians:
Army - MR
Navy – AS
Air Force - 20

Preparing activity:
Navy - AS

(Project 8030-2025-002)

Review activity:
Army - AV

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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